

Package name	Description	Remarks
1. Redfish	- Provides basic connectivity to HDFS - Performs reads/writes and modifies the files stored in HDFS - Is able to do file read/write and manipulation	- It depends on rJava - HADOOP_CMD environment variable is required to access HDFS
2. rhbase	- Provides basic connectivity to HBASE distributed databases - Performs read/write operations and modifies the tables stored in HBASE - Able to do table manipulation	Need to install and build Thrift.
3. plymr	- Performs common manipulation operations - Provides powerful ways to group the data	- It relies on Hadoop MapReduce to perform its tasks - It needs to be installed on each node of a Hadoop cluster together with its dependencies
4. rmr2	Performs statistical analysis via Hadoop MapReduce	It needs to be installed on each node in the cluster
5. ravro	Allows reading and writing of files in Avro format	Packages must be installed: Rcpp, rjson, bit64 and bit

Table 1: Packages of Hadoop

```
cat /etc/profile.d/hadoop.sh | sed 's/
export //g' > ~/.Renviron
wget http://download2.rstudio.org/
rstudio-server-0.97.332-x86_64.rpm
sudo yum install --nogpgcheck rstudio-
server-0.97.332-x86_64.rpm
```

- vii Install R dependencies.
- viii. Download and install rmr2.

The ORCH package

ORCH stands for 'Oracle R connector for Hadoop'. It consists of several packages that provide access to a Hadoop cluster, to achieve manipulation of HDFS resident data and the execution of MapReduce jobs. It consists of functions that access Apache Hadoop from a local R client. A data frame is the primary object type of ORCH. However, it works with vectors and matrices for the exchange of the data with HDFS.

In ORCH, input files need to satisfy several requirements in order to access HDFS files. These files must be in the 'comma separated' value

format, and an underscore as a first character is ignored. Also, all files for the MapReduce job must be stored in one directory as a part of a single logical file. ORCH also provides the preparation and analysis features of the Hive Query language. It accesses any Hive table containing the columns with string and numeric types such as integers, small integers, tiny integers, big integers, float and double.

The RHIPE package

RHIPE is another package that provides an API to use Hadoop with R for Big

Data analytics. It is more integrated with plots. RHIPE has several advantages such as:

- Large data visualisation
 - Handles large data sets
 - Is user friendly
 - Applies statistical algorithms
- To install RHIPE, follow the steps given below:
- i. Install Hadoop on clusters.
 - ii. Install R as a shared library.
 - iii. Install Google Protocol Buffers.
 - iv. Set environment variables like Hadoop and R Paths.
 - v. Install the RHIPE package. **END** 

References

- [1] https://en.wikipedia.org/wiki/Programming_with_Big_Data_in_R
- [2] <https://github.com/RevolutionAnalytics/RHadoop/wiki>
- [3] <http://deltarho.org/docs-RHIPE/>
- [4] https://subscription.packtpub.com/book/big_data_and_business_intelligence/9781782163282/3/ch03lvl1sec29/introducing-rhipe
- [5] https://docs.oracle.com/cd/E37231_01/doc.20/e36961/orch.htm#BDCUG216

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